

EGARCH(1,1,1) Model

Portfolio VaR Engine — Model Documentation

Risk Modelling — Trade Republic

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1 Overview

The Exponential GARCH model (EGARCH; Nelson, 1991) models the *logarithm* of the conditional variance, rather than the level. This has two key advantages over GJR-GARCH: (1) positivity of σ_t^2 is guaranteed without parameter constraints, and (2) both the magnitude and the sign of past shocks affect future volatility, naturally capturing the leverage effect.

Engine key: 'egarch'

Sources:

- `risk_models/garch_models.py`, lines 16–35 (`garch_exp`)
- `risk_models/garch_models.py`, lines 58–131 (`calculate_vaR_garch`)
- `risk_analytics/risk_engines.py`, lines 135–152 (`engine dispatch`)

Library: `arch` (Kevin Sheppard)

2 Volatility Model

2.1 EGARCH(1,1,1) Specification

Let r_t denote the portfolio return at time t . The model assumes:

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \stackrel{i.i.d.}{\sim} D(0, 1) \quad (1)$$

The log-variance dynamics are:

$$\log \sigma_t^2 = \omega + \alpha \left(\frac{|\varepsilon_{t-1}|}{\sigma_{t-1}} - \mathbb{E} \left[\frac{|\varepsilon_{t-1}|}{\sigma_{t-1}} \right] \right) + \gamma \frac{\varepsilon_{t-1}}{\sigma_{t-1}} + \beta \log \sigma_{t-1}^2 \quad (2)$$

- ω : intercept of the log-variance equation
- $\alpha \geq 0$: magnitude effect (symmetric response to shock size)
- γ : sign (asymmetry/leverage) effect; $\gamma < 0$ implies negative shocks increase volatility more than positive shocks
- β : persistence in log-variance; $|\beta| < 1$ ensures stationarity
- $\mathbb{E}[|z_{t-1}|]$ is a normalisation constant that depends on D

Because the variance is modelled in logarithms, *no non-negativity constraints* are needed on ω , α , γ , or β .

2.2 Comparison with GJR-GARCH

Feature	GJR-GARCH	EGARCH
Variance modelled	Level σ_t^2	Log log σ_t^2
Parameter constraints	Yes ($\omega, \alpha, \beta \geq 0$)	None
Leverage effect	Indicator $\mathbb{I}_{\varepsilon < 0}$	Sign term γz_{t-1}
Stationarity condition	$\alpha + \frac{1}{2}\gamma + \beta < 1$	$ \beta < 1$

2.3 ARCH library call

```
arch_model(rets_rescaled, p=1, q=1, o=1, vol='EGARCH', dist=dist)
```

Parameters `p=1`, `q=1` set the ARCH and GARCH lag orders; `o=1` activates the asymmetric (leverage) term.

3 VaR and CVaR Calculation

The VaR and CVaR computation is **identical** to the GJR-GARCH engine, both calling `calculate_var_garch()` with `garch_engine='egarch'`. For completeness, the formulas are reproduced below.

3.1 One-Step Forecast

$$\hat{\mu}_{t+1} = \hat{\mu}, \quad \hat{\sigma}_{t+1}^2 = \text{forecast variance} \quad (3)$$

3.2 Parametric VaR (without FHS)

Value-at-Risk:

$$\widehat{\text{VaR}}_{\alpha} = -\hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \cdot q_{\alpha}^D \quad (4)$$

CVaR — Normal:

$$\widehat{\text{CVaR}}_{\alpha}^N = -\hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \cdot \frac{\phi(\Phi^{-1}(\alpha))}{\alpha} \quad (5)$$

CVaR — Student- t :

$$\widehat{\text{CVaR}}_{\alpha}^t = -\hat{\mu}_{t+1} - \frac{\hat{\sigma}_{t+1}}{\alpha(1-\hat{\nu})} \left(\hat{\nu} - 2 + [t^{-1}(\alpha; \hat{\nu})]^2 \right) \cdot f_t(t^{-1}(\alpha; \hat{\nu}); \hat{\nu}) \quad (6)$$

3.3 Filtered Historical Simulation (FHS)

When `fhs=True`, standardised residuals $\hat{z}_t = (r_t - \hat{\mu})/\hat{\sigma}_t$ replace the parametric distribution quantile:

$$q_{\alpha}^{\text{FHS}} = Q_{1-\alpha}(\{\hat{z}_t\}) \quad (7)$$

3.4 Holding-Period Scaling

$$\widehat{\text{VaR}}_{\alpha}^{(H)} = \widehat{\text{VaR}}_{\alpha}^{(1)} \cdot \sqrt{H} \quad (8)$$

4 Parameters

Parameter	Type	Default	Description
window	int	250	Lookback window
rescale	float	1000	Rescaling factor for numerical stability
decay	float	1	EWMA pre-weighting decay ($\lambda = 1$: no decay)
ftol	float	1e-6	Optimizer convergence tolerance
max_iter	int	150	Maximum optimizer iterations
fhs	bool	False	Use Filtered Historical Simulation
distr	str	'Normal'	Innovation distribution
qtls	list	[0.05, 0.01, 0.001]	Tail probability levels
holding_period	int	1	Forecast horizon in days

5 Advantages and Limitations

Advantages	Limitations
No non-negativity constraints on parameters	More complex likelihood than GJR
Natural leverage effect via sign term	Harder to interpret parameters
Log-variance avoids negative variance	Sensitive to initialisation
Well-suited to equity return series	Square-root scaling is approximate

6 Usage Example

```
from risk_analytics import risk_engines as re

var_egarch = re.portfolio_var(
    rets_orig = rets,
    wgtts     = weights,
    engine     = 'egarch',
    fmt_engine = {
        'window'       : 250,
        'rescale'       : 1000,
        'decay'         : 1,
        'ftol'          : 1e-6,
        'max_iter'      : 150,
        'fhs'           : False,
        'distr'          : 't',
        'qtls'           : [0.05, 0.01, 0.001],
        'holding_period' : 1,
    }
)
```

7 References

Nelson, D.B. (1991). *Conditional Heteroskedasticity in Asset Returns: A New Approach*. *Econometrica*, 59(2), 347–370.